

# Lubomír Bureš

Nuclear engineer specialising in computational modelling and technical leadership.

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## Professional Experience

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### Saltfoss Energy (formerly Seaborg Technologies)

*Copenhagen, Denmark*

*Senior Multiphysics Specialist*

March 2024 – present

- Serve as a system technical lead within molten-salt reactor development, guiding design work for the system and coordinating with other system leads.
- Establish modelling approaches and develop tools for engineering analysis, then support colleagues in taking ownership while remaining available for technical review and guidance.
- Combine engineering leadership with hands-on modelling and analysis across reactor physics and thermal hydraulics, including development with the MOOSE Framework.
- Publish selected research and present technical developments at scientific conferences and industry forums.

*Multiphysics Specialist*

October 2021 – March 2024

- Developed and applied Python and Modelica models for molten-salt reactor physics, thermal hydraulics, and system dynamics.
- Supervised a master's thesis project whose candidate subsequently joined the company as a full-time employee.
- Published peer-reviewed research and represented the company both in Denmark and internationally.

### Paul Scherrer Institute

*Villigen, Switzerland*

*Doctoral Researcher*

October 2018 – September 2021

- Developed numerical methods and high-fidelity simulations of phase change and nucleate boiling.
- Implemented numerical methods in the C++ PSI-BOIL research code; verified and validated models and investigated microlayer dynamics.
- Supervised four student projects, resulting in published research.
- Led exercise sessions for a course at ETH Zurich.

## Supplementary Projects

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### Winterlead

*Remote*

*Independent Consultant*

April 2024 – present

- Completed projects in software development and computational engineering, including software for Quantified Carbon and volunteer technical contributions to the Repower Initiative.

## Education

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### EPF Lausanne, Institute of Physics

*Paul Scherrer Institute (PSI), Villigen, Switzerland*

*Doctor of Science (PhD)*

2018 – 2021

- *Thesis:* Fundamental study on microlayer dynamics in nucleate boiling
- *Achievements:* Awarded the 2022 PhD Prize by the European Nuclear Education Network Association.

**ETH Zurich – EPF Lausanne***Zürich & Lausanne, Switzerland**Joint MSc in Nuclear Engineering*

2016 – 2018

- *Achievements:* Graduated with distinction. Awarded the EPFL Excellence Fellowship over the duration of the study programme.
- *Certifications:* Received the European Master of Science in Nuclear Engineering certification.

**Czech Technical University***Prague, Czech Republic**BSc in Nuclear Engineering*

2013 – 2016

- *Achievements:* Graduated with honours.

**Skills**

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**Nuclear engineering:** Reactor physics, thermal hydraulics, molten-salt reactor systems  
**Computational modelling:** Multiphysics simulation, numerical methods, model verification and validation  
**Languages and frameworks:** Python, Modelica, MOOSE Framework

**Languages**

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Czech (native), English (C2), German (C1), Danish (B2)

**Organisational Involvement**

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**IDA Nuclear***Danish Society of Engineers, Denmark**Board Member*

September 2025 – present

- Organise events promoting knowledge and education about nuclear and atomic technology.

**Kildevæld Kulturcenter, Digitalt Værksted***Copenhagen, Denmark**Volunteer / Superuser*

October 2025 – present

- Support open workshop sessions by helping citizens use digital fabrication tools, including 3D printing and laser cutting, and by discussing and developing project ideas.

**PhD and Postdoc Association***Paul Scherrer Institute, Switzerland**President*

October 2019 – May 2021

- Led the association through the Covid-19 pandemic, oversaw bylaws enabling non-profit registration, and represented members within and outside PSI.
- March 2019 to September 2019: Social Committee Head.

**Various***Europe-wide**Volunteer / Event Organiser*

October 2018 – November 2023

- Volunteered for and helped organise multiple climate-change-related events across Europe.
- Represented the Nuclear for Climate initiative at the UN Climate Change Conference COP25 in Madrid.